

## PA44 Performance Calculations

Date: \_\_\_\_/\_\_\_\_/\_\_\_\_

Callsign: VH-\_\_\_\_

Route: \_\_\_\_\_

| <i>Aerodrome</i>          |  |  |  |
|---------------------------|--|--|--|
| <i>T (°C) / QNH</i>       |  |  |  |
| <i>Pressure Height</i>    |  |  |  |
| <i>ASDA</i>               |  |  |  |
| <i>TODR</i>               |  |  |  |
| <i>LDR</i>                |  |  |  |
| <i>SE Climb (FPM)</i>     |  |  |  |
| <i>SE Climb (%)</i>       |  |  |  |
| <i>SE Service Ceiling</i> |  |  |  |
| <i>ME Climb: Gear Dn</i>  |  |  |  |
| <i>ME Climb: Gear Up</i>  |  |  |  |

|                                | Weight (lbs) | Arm (in) | Moment (in-lbs) |
|--------------------------------|--------------|----------|-----------------|
| <i>BEW</i>                     |              |          |                 |
| <i>Row 1</i>                   |              | 80.5     |                 |
| <i>Row 2</i>                   |              | 118.1    |                 |
| <i>Bagg. (200 lbs Max)</i>     |              | 142.8    |                 |
| <i>ZFW</i>                     |              |          |                 |
| <i>Fuel (639 lbs Max)</i>      |              | 95.0     |                 |
| <i>Ramp Weight</i>             |              |          |                 |
| <i>Taxi Allowance (10 lbs)</i> | -10.0        | 95.0     | -950            |
| <i>TOW</i>                     |              |          |                 |

**MTOW:** 3800 lbs (1723 kgs)

**MLW:** 3800 lbs (1723 kgs)

**MRW (CAE):** 3810 lbs (1728 kgs)

**MRW:** 3816 lbs (1730 kgs)

**TOTAL FUEL:** 110 USG (416L)

**USABLE FUEL:** 108 USG (408L)

| PA44 SPECIFICATIONS     |                                                            |
|-------------------------|------------------------------------------------------------|
| <b>V<sub>SB</sub></b>   | 55 KIAS                                                    |
| <b>V<sub>S1</sub></b>   | 57 KIAS                                                    |
| <b>V<sub>MCA</sub></b>  | 56 KIAS                                                    |
| <b>V<sub>A</sub></b>    | 135 KIAS (3800 lbs/1725 kg)<br>112 KIAS (2700 lbs/1226 kg) |
| <b>V<sub>LO</sub></b>   | 109 KIAS                                                   |
| <b>V<sub>LE</sub></b>   | 140 KIAS                                                   |
| <b>V<sub>FE</sub></b>   | 111 KIAS                                                   |
| <b>V<sub>NO</sub></b>   | 169 KIAS                                                   |
| <b>V<sub>NE</sub></b>   | 202 KIAS                                                   |
| <b>V<sub>XWIN</sub></b> | 17 KIAS                                                    |
| <b>V<sub>R</sub></b>    | 75 KIAS (Normal)                                           |
| <b>V<sub>X</sub></b>    | 82 KIAS (SE and ME)                                        |
| <b>V<sub>Y</sub></b>    | 88 KIAS (SE and ME)                                        |
| <b>V<sub>CRZ</sub></b>  | 105 KIAS                                                   |

| FUEL CALCULATION | Min | L or KG | Min | L or KG | Min | L or KG |
|------------------|-----|---------|-----|---------|-----|---------|
| Trip Fuel        |     |         |     |         |     |         |
| Alternate        |     |         |     |         |     |         |
| Airwork          |     |         |     |         |     |         |
| Sub-Total        |     |         |     |         |     |         |
| Contingency Fuel |     |         |     |         |     |         |
| Final Reserve    |     |         |     |         |     |         |
| Holding Fuel     |     |         |     |         |     |         |
| Additional Fuel  |     |         |     |         |     |         |
| Taxi Fuel        |     |         |     |         |     |         |
| Fuel Required    | 1   | 2       |     |         |     |         |
| Fuel Margin      | 5   | 4       |     |         |     |         |
| Endurance        | 6   | 3       |     |         |     |         |
| From             |     |         |     |         |     |         |

| FUEL CALCULATION | Min | L or KG | Min | L or KG | Min | L or KG |
|------------------|-----|---------|-----|---------|-----|---------|
| Trip Fuel        |     |         |     |         |     |         |
| Alternate        |     |         |     |         |     |         |
| Airwork          |     |         |     |         |     |         |
| Sub-Total        |     |         |     |         |     |         |
| Contingency Fuel |     |         |     |         |     |         |
| Final Reserve    |     |         |     |         |     |         |
| Holding Fuel     |     |         |     |         |     |         |
| Additional Fuel  |     |         |     |         |     |         |
| Taxi Fuel        |     |         |     |         |     |         |
| Fuel Required    | 1   | 2       |     |         |     |         |
| Fuel Margin      | 5   | 4       |     |         |     |         |
| Endurance        | 6   | 3       |     |         |     |         |
| From             |     |         |     |         |     |         |

Fuel flow Trip – 85 L/hr

Holding – 60 L/hr

Taxi Fuel – 8 L per full stop landing

Heater Operation – 2 L/hr (left tank)